European Polymer Journal 179 (2022) 111520

Contents lists available at ScienceDirect

European Polymer Journal

journal homepage: www.elsevier.com/locate/europolj

Thermally latent vinyl crosslinking of polymers via sulfoxide chemistry

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ARTICLE INFO

Keywords:
Polymer crosslinking
Latent crosslinking
Vinyl protection
Sulfoxide chemistry

ABSTRACT

Radical crosslinking reactions utilizing vinyl functionality are widely used to fabricate polymeric networks for bulk and surface applications. Here, we report a thermally latent vinyl crosslinking of polymers involving sulfoxide chemistry for the first time. Polymers with sulfoxide-protected methacrylate moieties are prepared via free-radical polymerization. These moieties are deprotected by a thermal process and transformed into methacrylate functional groups. The crosslinking behaviors of the protected polymers are investigated using an oscillatory rheometer in the presence of a radical initiator. The behavior strongly depended on the formation of methacrylate species in polymer chains. The thermally latent crosslinking approach introduces a novel method for the stepwise and systematic construction of polymeric networks that can potentially be applied in dual-curing or patterning processes.

1. Introduction

Chemical crosslinking of polymers improves their mechanical, thermal, and physicochemical properties by forming a network structure from polymer chains.[1–6] Crosslinking of polymers with vinyl functional groups is one of the most studied areas in polymer chemistry due to their high selectivity and reactivity toward radicals.[7–9] The radical crosslinking reactions generally can be triggered by light and heat.[10] Light-induced crosslinking can achieve fast curing at room temperature [11] and has been recognized as an eco-friendly process due to the low energy consumption and emission levels of volatile organic compounds. [12–15] However, these light-induced radical generation processes are mainly used for surface networks because of the penetration depth of light. On the other hand, thermal-induced crosslinking is one of the most used because it has fewer restrictions on the material's volume or thickness and is easier to implement than the former. For this reason, thermal-induced crosslinking has been extensively utilized to fabricate polymeric materials from the surface (i.e., coatings and films) to bulk (i.e., rubbers, thermoset elastomers, gels, and particles) network structures.[2,16].

The thermal-induced radical crosslinking can be controlled by varying the radical concentration or the number of vinyl groups. The former has been performed by thermal radical initiators according to their half-life. Meanwhile, controlling the number of vinyl groups in polymer chains, especially during crosslinking, has rarely been realized. A possible approach is the protection/deprotection of the double bond; a representative example is furan-protected maleimide chemistry [17,18] which has been used in polymeric conjugations [19,20] and self-healings.[21] However, the furan-protected maleimide chemistry is disadvantageous for homopolymerization and requires using comonomers. [22] These problems can be mitigated by utilizing sulfoxide-protected vinyl chemistry, in which a sulfoxide group becomes deprotected and is transformed into a vinyl group by a thermal process.[23] The kinetics of sulfoxide deprotection in the various vinyl models, including acrylate, methacrylate, and styrene, were investigated in our previous study.[24].

Herein, we report a new approach to controlling vinyl functionality via sulfoxide chemistry during a thermal crosslinking reaction for the first time (Scheme 1). Functional polymers with sulfoxide-protected methacrylate (SMA) moiety were prepared by free-radical polymerization. Upon thermal process, the SMA moiety in the polymers was deprotected into a methacrylate group for crosslinking reactions. The evolution of the crosslinking behavior was monitored in the presence of a thermal initiator. The sulfoxide-based vinyl protection enabled a thermally latent crosslinking and thus can potentially be applied in

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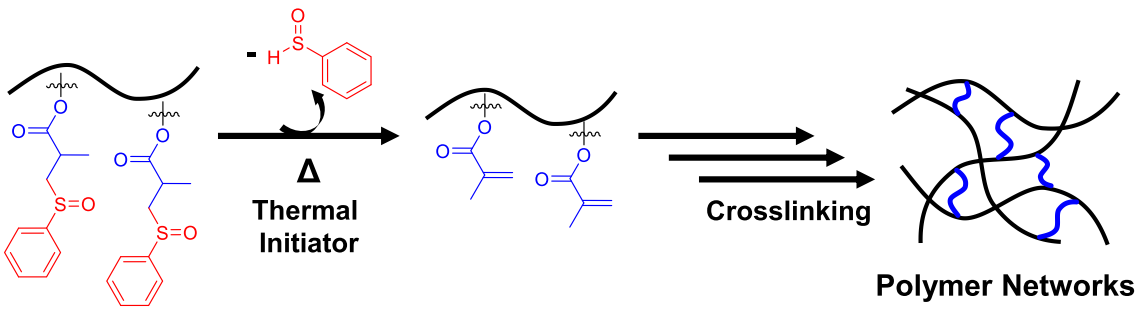
https://doi.org/10.1016/j.eurpolymj.2022.111520
Received 14 June 2022; Received in revised form 12 August 2022; Accepted 16 August 2022
Available online 22 August 2022
0014-3057/© 2022 Published by Elsevier Ltd.



Scheme 1. Schematic overview of the thermally latent crosslinking reaction through controlling vinyl functionality via sulfoxide chemistry.

thermal dual-curing (e.g., thermal-thermal or photo-thermal system) or patterning processes. This approach allows for the stepwise and systematic construction of polymer networks and may improve the final product's physical properties, appearance, and processability.

2. Experimental

2.1. Materials

2-hydroxyethyl methacrylate (HEMA, TCI, 97.0%), methyl methacrylate (MA, TCI, 99.0%), n-butyl acrylate (BA, Sigma Aldrich, 99.0%), and styrene (St, DAEJUNG, 99.0%) were purified by passing through a basic alumina column before use to remove radical inhibitor. Thiophenol (Sigma Aldrich, 97.0%), triethylamine (TEA, DUKSAN, 98.0%), tetrahydrofuran (THF, DUKSAN, 99.9%), acryloyl chloride (TCI, 97.0%), acetic acid (Sigma Aldrich, 99.9%), sodium hydroxide (NaOH, DUKSAN, 93.0%), methylene chloride (CH_2Cl_2, SAMCHUN, 99.5%), ethyl acetate (EtOAc, DUKSAN, 99.5%), hexane (DUKSAN, 95.0%), hydrogen peroxide solution 30% (w/w) in water (H_2O_2, Sigma-Aldrich), sodium carbonate anhydrous (Na_2CO_3, DAEJUNG, 99.5%), magnesium sulfate ($MgSO_4$, DAEJUNG, 99.5%), 2,2′-azobis(2,4-dimethylvaleronitrile), (V-65, Wako Chemical, 95.0%), *tert*-dodecylmercaptan (TDM, Sigma Aldrich, 98.0%), ethyl ether (DAEJUNG, 99.5%), methacryloyl chloride (Alfa Aesar, 97.0%), anisole (Sigma Aldrich, 99.9%), t-amyl peroxy-2-ethylhexanoate (Nouryon, 99.0%), t-butyl peroxide (Sigma Aldrich, 98.0%), and t-butyl peroxybenzoate (Sigma Aldrich, 98.0%) were used as received.

2.2. Synthesis of SMA-functionalized ethyl acrylate

2-hydroxyethyl 2-methyl-3-(phenylthio)propanoate. HEMA (4.00 g, 30.7 mmol), thiophenol (6.80 g, 61.5 mmol), TEA (6.20 g, 61.5 mmol), and dry THF (100 mL) were added to a 250 mL round flask and stirred for 48 h at room temperature. The reaction mixture was quenched with H_2O and extracted with EtOAc. The organic layer was separated, dried over $MgSO_4$, and concentrated. The residue was subjected to column chromatography with hexane/EtOAc (2/1) as eluent to give the desired product (7.00 g, 95%) as a colorless oil. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.39–7.21 (m, 5H), 4.29–4.19 (m, 2H), 3.84 (q, J = 7.7 Hz, 2H), 3.23 (dd, J = 16.0, 8.0 Hz, 1H), 3.01 (dd, J = 14.0, 4.0 Hz, 1H), 2.77 (quin, J = 7.0 Hz, 1H), 2.02 (t, J = 6.4 Hz, 1H), 1.30 (d, J = 7.0 Hz, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 175.2, 135.2, 129.9, 128.9, 126.5, 126.4, 65.9, 60.5, 39.4, 37.1, 16.5.

4-methyl-3-oxo-5-(phenylthio)pentyl acrylate. 2-hydroxyethyl 2-methyl-3-(phenylthio)propanoate (5.00 g, 20.8 mmol), TEA (8.42 g, 83.2 mmol), and dry THF (150 mL) were added to a 250 mL round flask. Acryloyl chloride (7.53 g, 83.2 mmol) was added dropwise to the flask at 0 °C over a period of 30 min. The solution was stirred for 2 h at room temperature. The reaction mixture was quenched with H_2O and extracted with EtOAc. The organic layer was separated, washed with saturated aq. Na_2CO_3 solution, dried over $MgSO_4$, and concentrated to give the desired product (6.10 g, 84%) as a colorless oil. The residue was used for the next step without any further purification. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.38–7.19 (m, 5H), 6.44 (d, J = 17.4 Hz, 1H), 6.14 (dd, J = 17.2, 10.3 Hz, 1H), 5.87 (d, J = 10.5 Hz, 1H), 4.38–4.31 (m, 4H), 3.26 (dd, J = 13.3, 7.3 Hz, 1H), 2.94 (dd, J = 13.5, 6.6 Hz, 1H), 2.73 (q, J = 7.3 Hz, 1H), 1.28 (d, J = 7.3 Hz, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 174.1, 165.3, 135.3, 130.8, 129.7, 128.5, 127.6, 126.1, 61.8, 61.7, 39.3, 36.9, 16.2.

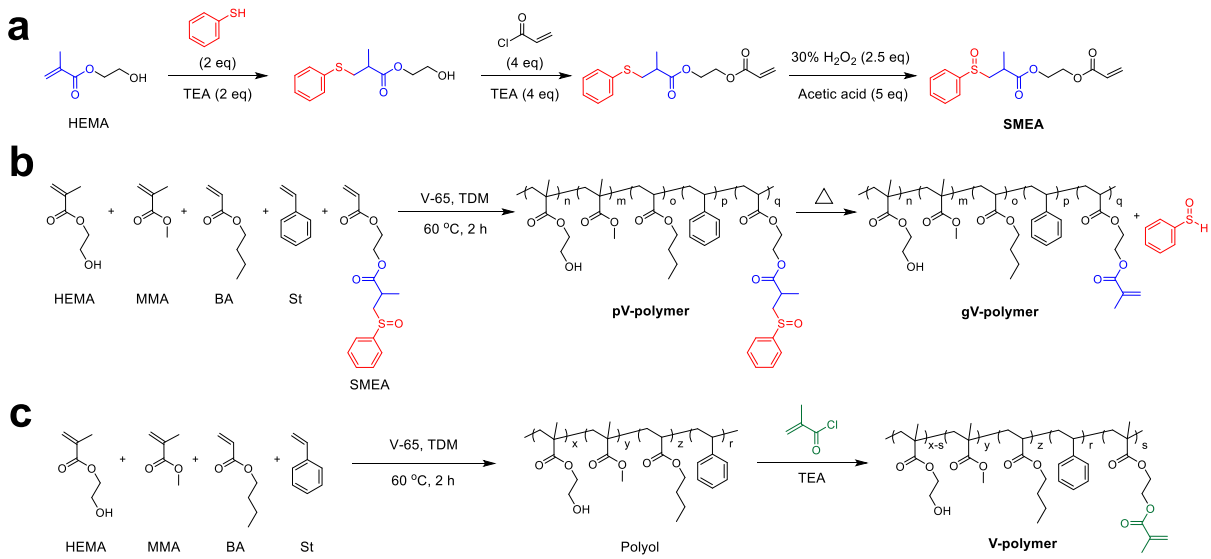
2-((2-methyl-3-(phenylsulfinyl)propanoyl)oxy)ethyl acrylate (SMEA). 4-methyl-3-oxo-5-(phenylthio)pentyl acrylate (3.50 g, 11.9 mmol), acetic acid (3.41 mL, 59.4 mmol), and CH_2Cl_2 (35 mL) were added to a 100 mL round flask. 30% aq·H_2O_2 solution (3.03 mL, 29.7 mmol) was added dropwise to the flask at room temperature. After stirring for 18 h at room temperature, the solution was cooled to 0 °C and aq. 4 M NaOH was added to the solution until it became to pH 10. Then, the reaction mixture was extracted with CH_2Cl_2. The organic layer was separated, dried over $MgSO_4$, and concentrated. The residue was subjected to column chromatography with hexane/EtOAc (1/2) as eluent to give the desired product (3.00 g, 81%) as a colorless oil. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.67–7.63 (m, 2H), 7.55–7.52 (m, 3H), 6.46–6.39 (m, 1H), 6.17–6.07 (m, 1H), 5.88–5.84 (m, 1H), 4.45–4.41 (m, 2H), 4.35–4.31 (m, 1H), 4.27–4.24 (m, 1H), 3.28–3.21 (m, 1H), 3.16–3.04 (m, 1H), 2.80–2.72 (m, 1H), 1.42 (d, J = 6.9 Hz, 1.5H), 1.31 (d, J = 6.8 Hz, 1.5H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 173.7, 165.6, 143.4, 143.1, 131.3, 131.1, 129.1, 127.7, 123.9, 123.8, 62.6, 61.8, 60.9, 60.0, 34.3, 33.7, 17.8, 16.6.

2.3. Preparation of crosslinkable polymer and polymer film

Vinyl protected polymer (pV-polymer). A degassed mixture of HEMA (5.00 g, 38.4 mmol), MMA (5.12 g, 51.2 mmol), BA (1.64 g, 12.8 mmol), St (1.33 g, 12.8 mmol), SMEA (3.97 g, 12.8 mmol), V-65 (4.77 g, 19.2 mmol), TDM (0.518 g, 2.56 mmol), and dry THF (25.2 mL) was added to the 100 mL round flask via N_2-purged syringes. The reaction mixture was placed in an oil bath at 60 °C under stirring for 2 h. The monomers conversion was obtained using ^{1}H NMR spectroscopy ($CDCl_3$, 25 °C). After the reaction, the mixture was precipitated twice in 2,000 mL of cold ethyl ether under stirring. The polymer was isolated by filtration and vacuum evaporation. A number-average molar mass ($M_{n,SEC}$) and molecular weight distribution ($Đ$) of pV-polymer were found to be 4.1 kg mol^{-1} and 1.47, respectively.

Vinyl generating polymer (gV-polymer). Into a 100 mL round flask, pV-polymer (10.0 g) and anisole (50.0 mL) were added. The mixture was stirred at 110 °C for 3 h. The solvent was removed by freeze-drying for 3 days. $M_{n,SEC}$ and $Đ$ of gV-polymer were found to be 4.1 kg mol^{-1} and 1.45, respectively.

OH functionalized polymer (polyol). A degassed mixture of HEMA (10.0 g, 76.8 mmol), MMA (7.69 g, 76.8 mmol), BA (2.46 g, 19.2 mmol), St (2.00 g, 19.2 mmol), V-65 (7.16 g, 28.8 mmol), TDM (0.777 g, 3.84 mmol), and dry THF (33.8 mL) was added to the 100 mL round flask via N_2-purged syringes. The reaction mixture was placed in an oil bath at



Scheme 2. Syntheses of (a) SMEA and (b and c) vinyl protected (pV-), vinyl generating (gV-), and vinyl functionalized (V-) polymer.

60 °C under stirring for 2 h. The monomers conversion was obtained using ^{1}H NMR spectroscopy. After the reaction, the mixture was precipitated twice in 2,000 mL of cold ethyl ether under stirring. The polymer was isolated by filtration and vacuum evaporation. $M_{n,SEC}$ and $Đ$ of polyol were found to be 4.2 kg mol^{-1} and 1.35, respectively.

Vinyl functionalized polymer (V-polymer). Polyol (15.0 g), methacryloyl chloride (1.55 g, 14.9 mmol), and dry THF (25.0 mL) were added to the 100 mL round flask and stirred for 30 min at 0 °C ice bath. TEA (1.50 g, 14.9 mmol) in dry THF (5.00 mL) was added dropwise into the flask. After 24 h stirring at room temperature, the mixture was precipitated twice in 2,000 mL of cold ethyl ether under stirring. The polymer was isolated by filtration and vacuum evaporation. The product was dissolved in CH_2Cl_2 (30.0 mL). After extracting with water (20.0 mL), the CH_2Cl_2 layer was dried at room temperature in a vacuum oven. $M_{n,SEC}$ and $Đ$ of V-polymer were found to be 4.3 kg mol^{-1} and 1.33, respectively.

Polymer film. The polymers were dissolved in anisole (50% w/w) with an equimolar of t-butyl peroxybenzoate to the sulfoxide or corresponding vinyl group of the polymers. The mixture was loaded into a rectangular-shaped Teflon mold (5.6 × 1.1 × 0.1 cm) and cured at 130 °C for 3 h after 5 min of flash off. The thermal properties of the films were characterized using DSC and TGA. These polymeric mixtures were coated onto tin substrates using the drawdown bar- coating method and cured under the same conditions to analyze the mechanical properties via a nano-scratch tester.

2.4. Characterization

Size exclusion chromatography (SEC). SEC experiment was carried out by an Agilent 1100 pump, a refractive index detector, PSS SDV precolumn (5 μm; 50.0 × 8.0 mm), and PSS SDV columns (5 μm; 10^5, 10^3, and 10^2 Å; 300.0 × 8.0 mm) with calibration using poly(methyl methacrylate) (PMMA) standards·THF was used as an eluent at 40 °C with a flow rate of 1.0 mL min^{-1}.

Nuclear magnetic resonance (NMR). ^{1}H NMR spectra were obtained on a 400 MHz Agilent Superconducting FT-NMR spectrometer using chloroform-d ($CDCl_3$) as the solvent.

Differential scanning calorimetry (DSC). The glass transition temperature (T_g) of polymers and crosslinked films were measured using a DSC (Q 2000, TA Instruments, USA). The heating and cooling temperature range were from −40 to 120 °C (1st cycle: heat up to 120 °C, cool to −40 °C at a rate of 10 °C min^{-1}, 2nd cycle: heat up to 120 °C at a rate of 5 °C min^{-1}) under a nitrogen atmosphere. Therefore, the T_g values were determined from 2nd heating run.

Thermal gravimetric analyzer (TGA). The thermal stability of the crosslinked films was investigated by thermogravimetric analysis (Q 500, TA Instruments, USA). The TGA measurements were performed from 30 to 600 °C at a heating rate of 10 °C min^{-1} under a nitrogen atmosphere to check the thermal properties at high temperatures. Typically, 5 wt% losses of the samples were considered the thermal degradation temperature (T_d).

Oscillatory Rheometer. The curing behaviors of the polymeric mixtures were characterized with by oscillatory rheometer (Thermo Fisher Scientific Inc., MARS III) operated at a constant frequency of 5 Hz and a strain of 0.1 % using small amplitude oscillatory shear (SAOS) mode at room temperature. These were placed between two parallel plates. A disposable plate with 25 mm diameter was used in this test. During the measurement, the gap between the upper and lower plates was 250 μm. The measurement temperature was increased from 30 to the target curing temperature of 110, 130, and 150 °C at a heating rate of 10 °C min^{-1} and then maintained for 2 h.

Nano-scratch tester (NST). The scratch penetration depths were observed by a NST (Anton Paar Tritec SA, Switzerland). During the tests, samples were scratched with a diamond sphero-conical tip with a radius of 2 μm. The vertical load force acting on the cured coating surface is progressively increased from 0.1 to 60 mN with a scanning speed of 1.5 mm min^{-1} and 1 mm in length. After recording the depth positions, we evaluated the scratch performances of the cured coating films.

3. Results and discussion

3.1. SMA-functionalized acrylate

Vinyl protection of the monomer was realized via temperature-dependent sulfoxide chemistry.[24] SMA-functionalized ethyl acrylate (SMEA) was prepared by modifying an acrylic monomer advantageous for polymerization at low-temperature (Scheme 2a). In this process, HEMA was thiolated using thiophenol by a 1,4-addition reaction. The hydroxyl group of the resulting product was reacted with acryloyl chloride, and then the oxidation of the sulfide group into the corresponding sulfoxide group readily produced SMEA. The structure of SMEA was characterized by ^{1}H and ^{13}C nuclear magnetic resonance (NMR) spectroscopy (Fig. S1a and b, respectively). SMEA was subsequently used as a comonomer to prepare crosslinkable polymers.

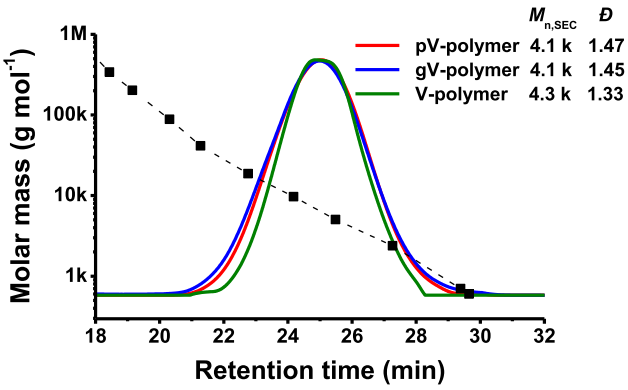


Fig. 1. Molecular weight comparison between pV-polymer (—, red line), gV-polymer (—, blue line), and V-polymer (—, green line) via SEC calibrated using PMMA standards (■, black filled square).

3.2. Crosslinkable polymers

The molar mass and Tg values of crosslinkable polymers were designed to have less than 6 kg mol^{-1} and 50 °C, respectively, considering the investigation of curing behavior involving potential dual curing systems.[11,13] We hypothesized that the precise generation of vinyl groups into polymer chains was a key to thermally latent vinyl crosslinking. Thus, vinyl-protected polymers (pV-polymer) were prepared via free-radical polymerization with a chain transfer agent under mild conditions (at 60 °C for 2 h) to suppress sulfoxide elimination reaction in synthetic steps (Scheme 2b). pV-polymer was synthesized using HEMA, MMA, BA, St, and SMEA as the monomers; V-65 as the initiator; and TDM as the chain transfer agent; ([HEMA]$_0$: [MMA]$_0$: [BA]$_0$: [St]$_0$: [SMEA]$_0$: [V-65]$_0$: [TDM]$_0$ = 30: 40: 10: 10: 10: 15: 2 in THF, [HEMA]$_0$ = 0.80 mM). The total monomer consumption determined by ^{1}H NMR ($CDCl_3$, 25 °C) reached 99% after 2 h. The resulting polymer was purified by the precipitation in cold diethyl ether followed by drying in vacuo. $M_{n,SEC}$ and $Đ$ of polymer were measured by the THF-eluted SEC, and their values determined using PMMA standards were 4.1 kg mol^{-1} and 1.47, respectively (Fig. 1, red line). The chemical structure of pV-polymer was characterized by ^{1}H NMR ($CDCl_3$, 25 °C; see Fig. 2a). The pV-polymer was composed of HEMA (b), MMA (c), BA (d), St (e), and SMEA (j) with a molar ratio of 31: 40: 10: 10: 9. Vinyl peaks between 5.6 and 6.1 ppm were not observed, indicating that the heat-labile sulfoxide groups remained unaffected during the polymerization process. Subsequently, a vinyl generating polymer (gV-polymer) was prepared by deprotecting the sulfoxide group of pV-polymer (Scheme 2b) and analyzed by ^{1}H NMR (Fig. S3). The solution of pV-polymer in anisole was heated to 110 °C for 3 h and isolated via freeze-drying to incorporate methacrylate functional groups into polymer chains. Vinyl generation was observed by ^{1}H NMR (δ 5.6 and 6.1 ppm, Fig. 2b), whereas the peaks corresponding to methine, methylene, and aromatic compounds (h, i, and j, respectively) disappeared. The composition of gV-polymer was unaffected during sulfoxide elimination, and the HEMA: MMA: BA: St: MEA = 31: 40: 10: 10: 9 ratio was preserved. Furthermore, $M_{n,SEC}$ and $Đ$ were equal to 4.1 kg mol^{-1} and 1.45, respectively (Fig. 1, blue). As a control experiment, a vinyl functionalized polymer (V-polymer) with similar vinyl contents and hydroxyl groups to gV-polymer was prepared. The reaction was conducted in THF at [HEMA]$_0$: [MMA]$_0$: [BA]$_0$: [St]$_0$: [V-65]$_0$: [TDM]$_0$ = 40: 40: 10: 10: 15: 2, a temperature of 60 °C, and [HEMA]$_0$ = 1.18 mM. The higher [HEMA]$_0$ concentration as compared with that used in the reaction of the pV-polymer was required to utilize hydroxyl groups for vinyl functionalization. After the isolation by precipitation, methacrylic groups were introduced via nucleophilic substitution under basic conditions (Scheme 2c). The resulting polymer was purified by the precipitation in cold diethyl ether followed by drying in vacuo, and its $M_{n,SEC}$ and $Đ$ values were equal to 4.3 kg mol^{-1} and 1.33, respectively (Fig. 1, green line). The polymer composition corresponded to the molar ratio HEMA: MMA: BA: St: MEA = 31: 40: 10: 10: 9, which was similar to that of gV-polymer determined by ^{1}H NMR (Fig. 2c). Therefore, the synthesized

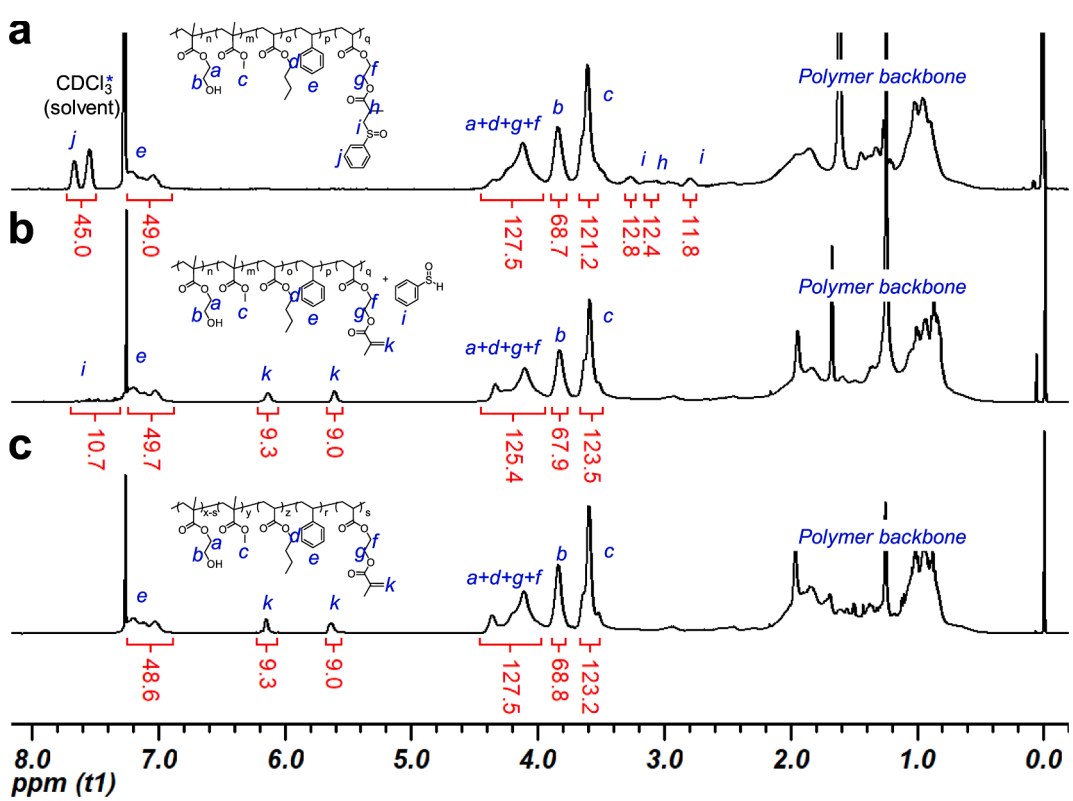


Fig. 2. Molecular structures of (a) pV-polymer, (b) gV-polymer, and (c) V-polymer analyzed by ^{1}H NMR ($CDCl_3$, 25 °C).

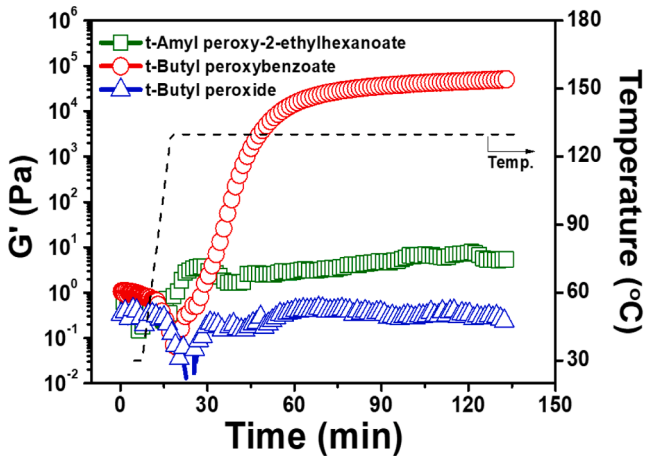


Fig. 3. Effect of the half-life of the thermal initiators (t-amyl peroxy-2-ethylhexanoate (□), t-butyl peroxybenzoate (○), and t-butyl peroxide (△)) on the storage modulus (G') of pV-polymer during crosslinking process studied using oscillatory rheometer.

three polymers exhibited similar $M_{n,SEC}$ values and monomer composition, except vinyl functionality.

3.3. Thermal crosslinking of the synthesized polymers

As a preliminary curing experiment, the effects of various thermal initiators (t-amyl peroxy-2-ethylhexanoate, t-butyl peroxybenzoate, and t-butyl peroxide with 1 h half-lives at 92, 124, and 149 °C, respectively) on the pV-polymer properties were investigated. The reaction was monitored by an oscillatory rheometer for 130 min at 130 °C because the yield of the generated vinyl groups in pV-polymer exceeded 99% under such conditions.[24] Thus, a crosslinking reaction between the initiator and vinyl-containing polymer can occur. We observed the growth of the crosslinked network by monitoring the time change of the storage modulus (G') (Fig. 3). For the initiator with a relatively faster or slower radical initiation rate (t-amyl peroxy-2-ethylhexanoate or t-butyl peroxide, respectively) than vinyl generation rate, minor changes in G' were observed. In contrast, the reaction with t-butyl peroxybenzoate demonstrated increasing G' values. After considering both the rate of vinyl generation and radical initiation, we concluded that t-butyl peroxybenzoate effectively promoted the crosslinking reaction.

Next, the thermal curing behaviors of the synthesized polymers were investigated at various temperatures (110, 130, and 150 °C) using t-butyl peroxybenzoate as the thermal initiator. The onset time of pV-polymer at 110 °C was 64 min, while the corresponding values obtained for gV-polymer and V-polymer were 23 and 13 min, respectively (Fig. 4a). Similarly, the crosslinked pV-polymer network grew much slower than those of gV-polymer and V-polymer. At a curing temperature of 130 °C, the onset time of pV-polymer decreased to 20 min, and the crosslinking density gradually increased (Fig. 4b). Finally, the G' values reached the levels obtained for the other polymers after 90 min. The vinyl generation of pV-polymer at 130 °C proceeded much faster than the curing reaction at 110 °C. However, at 150 °C, all polymers exhibited similar curing characteristics, and the onset times of pV-polymer, gV-polymer, and V-polymer were 17, 15, and 13 min, respectively (Fig. 4c). This phenomenon was observed due to the rapid vinyl generation from SMA moieties of pV-polymer and fast crosslinking reaction. Considering the G' values obtained crosslinking process, we concluded that by varying the onset time of pV-polymer, its curing behavior could be effectively tuned. As a result, the thermally latent vinyl crosslinking reaction was realized for the polymer containing SMA moieties (pV-polymer) at various curing temperatures.

The thermal properties of the synthesized polymer films after curing were examined by DSC and thermos gravimetric analysis (TGA) and

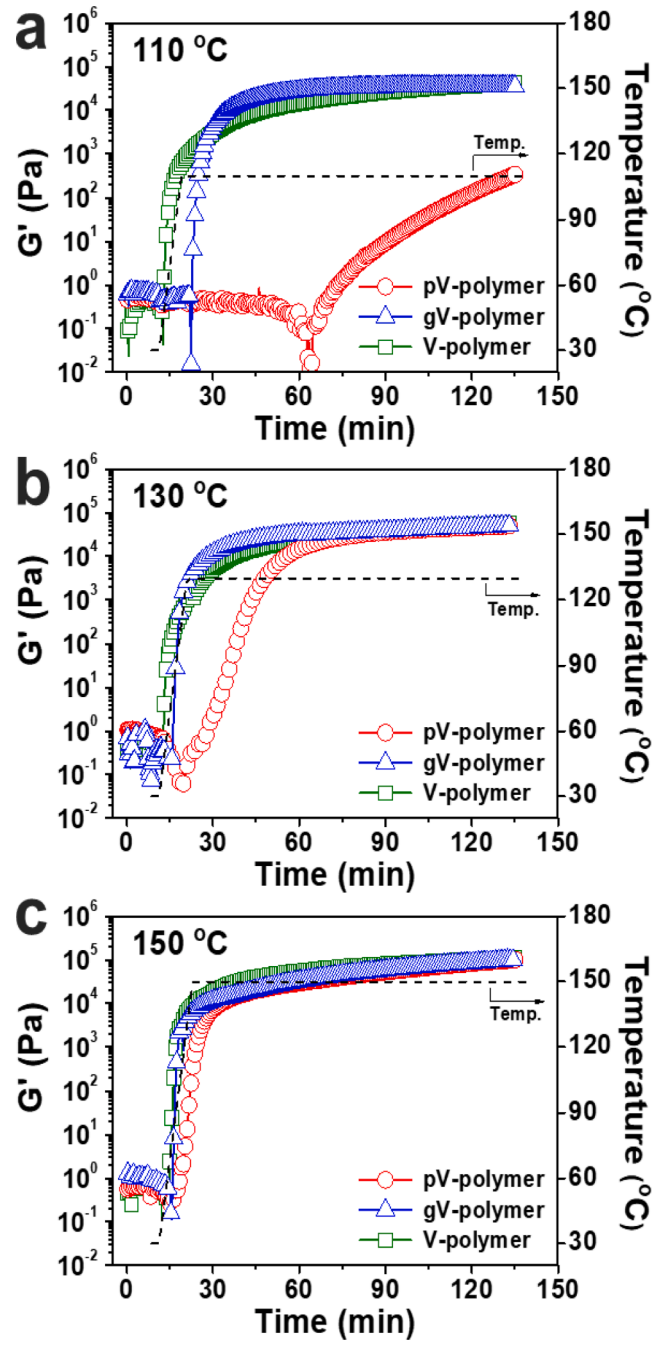


Fig. 4. Curing behaviors of the synthesized polymers (pV- (○), gV- (△), and V- (□) polymer) at (a) 110 °C, (b) 130 °C, and (c) 150 °C measured by the oscillatory rheometer.

compared with those of the corresponding bulk polymers. The films were prepared from the mixtures of t-butyl peroxybenzoate with the polymers in anisole cured at 130 °C for 3 h. The T_g of pV-film, gV-film, V-film determined by DSC were 65.0, 65.4, and 65.0 °C (Fig. 5b), while the corresponding values obtained for the bulk polymers were equal to 43.4, 42.5, and 49.7 °C, respectively (Fig. 5a). After the film preparation, the T_g of the polymer increased due to the formation of crosslinked networks. The thermal stability of the films was investigated by TGA in the temperature range between 30 and 600 °C. The temperatures, at which 5 wt% losses were observed for pV-film, gV-film, and V-film (T_d), were equal to 218.4, 218.4, and 248.2 °C (Fig. 5d), while the corresponding values obtained for the bulk polymers were 194.7, 214.2, and 234.5 °C, respectively (Fig. 5c). Hence, the curing process considerably enhanced

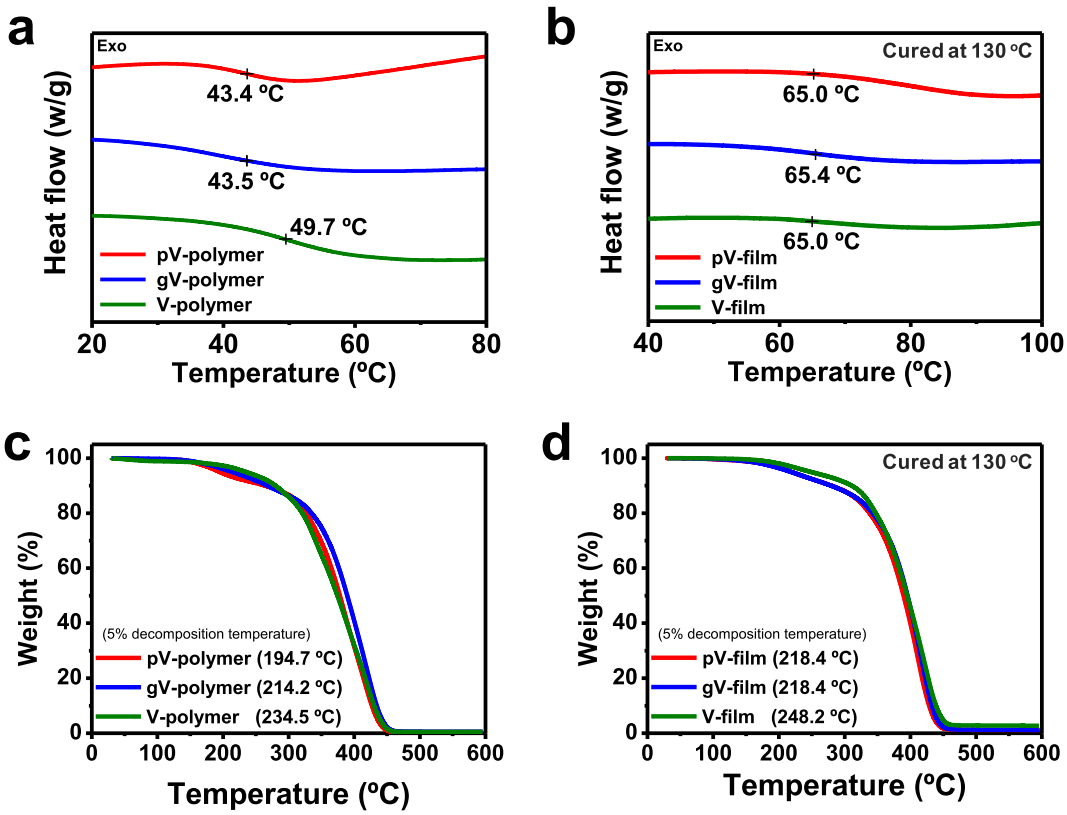


Fig. 5. Thermal analysis of crosslinkable polymers and corresponding films. Glass transition temperature measured by DSC for (a) polymers and (b) polymeric films cured at 130 °C; thermal stability by TGA for (c) polymers and (d) polymeric films cured at 130 °C.

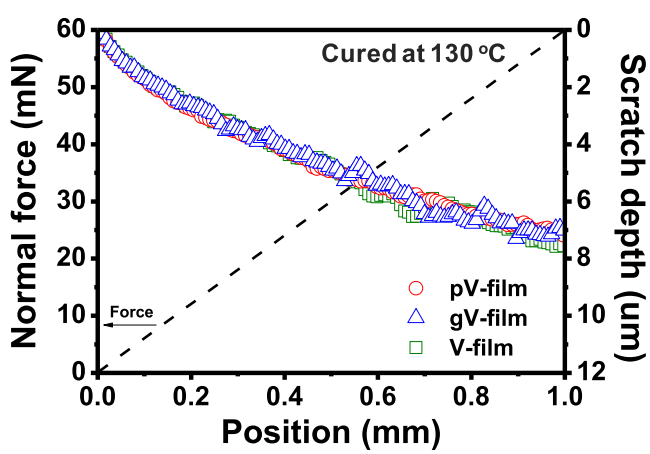


Fig. 6. Penetration depths of pV-film (○), gV-film (△), and V-film (□) measured by the nano-scratch tester.

the polymer thermal properties due to crosslinking networks.

Scratch penetration depths and patterns of the polymer films were measured by a nano-scratch tester to evaluate their scratch resistance properties (Fig. 6). The applied normal force gradually increased from 0.1 to 60 mN to obtain the scratch–depth profiles of the films. All films exhibited similar penetration depth profiles, which strongly correlated with the final modulus values of the polymers cured at 130 °C (Fig. 4b). Therefore, the polymer curing reactions were complete under the specified conditions. Based on the curing kinetics of polymers and the final mechanical properties of the corresponding polymeric films, a novel thermal latent vinyl crosslinking reaction of sulfoxide-protected polymers may contribute to developing dual cure systems.

4. Conclusions

The controlling vinyl functionality via sulfoxide chemistry was applied to the polymer crosslinking process. For this purpose, SMA-functionalized acrylate was synthesized and then employed as a monomer to prepare the protected polymers. Upon the thermal process of the protected polymers, methacrylate-functionalized polymers were obtained at relatively high yields determined by ^{1}H NMR. The polymer crosslinking reactions were systematically investigated using an oscillatory rheometer under different temperatures. During the reaction, the storage modulus of the protected polymers gradually increased, whereas a rapid increase was observed in the methacrylate-functionalized polymers. These results indicate that the thermal process leads to the subsequent crosslinking reaction by the methacrylate generation. In addition, the thermal and scratch resistance properties of protected polymers have been shown to exhibit similar values without deterioration when compared to methacrylate-functionalized polymers. Therefore, thermal latent crosslinking has been successfully realized using functional polymers with sulfoxide-protected vinyl moieties. The obtained polymer template can be potentially used to control the vinyl crosslinking rate in dual-curing systems.

CRediT authorship contribution statement

Dongwoo Kim: Data curation, Formal analysis, Investigation, Methodology, Writing – original draft. **Sunghyun Kim:** Data curation, Methodology, Writing – original draft. **Seokhyeon Jeong:** Formal analysis. **Minji Kim:** Investigation. **Woong Ki Hong:** Data curation, Methodology. **Heung Bae Jeon:** Investigation, Validation. **Hong Y. Cho:** Funding acquisition, Writing – review & editing, Supervision. **Seung Man Noh:** Funding acquisition, Conceptualization, Project administration, Supervision. **Hyun-jong Paik:** Funding acquisition, Conceptualization, Project administration, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work was supported by the Materials and Components Technology Development Program (20010566 and 20010256) grant funded by the Ministry of Trade, Industry, and Energy of Korea (MOTIE), Korea Research Institute of Chemical Technology (KRICT) Research and Development (R&D, SS2241-10) program, and Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (2022R1I1A1A01071904). H.B. Jeon thanks the research grant from Kwangwoon University in 2022. D. Kim and S. Kim contributed equally to this work.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2022.111520.

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